

SHIKHAR TIWARI

AI & Industrial Automation Engineer | Industry 4.0 • IIoT • Predictive Maintenance

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PROFESSIONAL SUMMARY

Software engineer with 12+ years of experience building scalable, data-intensive applications, now focused on Industry 4.0 and Industrial IIoT. Currently automating shop-floor operations by integrating PLC and SCADA systems and building the IIoT data pipelines and Manufacturing Execution System (MES) that link factory equipment to enterprise ERP and the cloud. Brings a strong backend, data-engineering, and cloud foundation to industrial AI, and is actively developing machine-learning skills for predictive maintenance, anomaly detection, and digital-twin applications. Motivated by turning real-time machine and sensor data into intelligent, self-optimizing manufacturing systems.

TECHNICAL SKILLS

Industry 4.0 & Automation: IIoT architecture, PLC, SCADA, MES development, ERP integration, OT/IT convergence, shop-floor data acquisition, real-time monitoring dashboards

Industrial Data & AI (developing): Time-series & sensor data, data pipelines / ETL, predictive maintenance, anomaly detection, digital twin, foundational machine learning

Backend & APIs: Node.js, Express.js, REST APIs, microservices architecture, real-time systems (Socket.IO / MQTT)

Databases: MongoDB, MySQL, NoSQL & schema design, time-series data modelling

Cloud & DevOps: AWS (EC2, S3, Lambda), CI/CD pipelines, cloud deployment

Frontend & Dashboards: React.js, Next.js, TypeScript, Flutter — operator/HMI-style dashboards & analytics UIs

Programming Languages: JavaScript, TypeScript, Python (developing), Dart.

PROFESSIONAL EXPERIENCE

Senior Software Engineer → Industry 4.0 / IIoT

Mar 2022 – Present

Nortcele Systems Pvt. Ltd., Bengaluru

Industry 4.0 & Industrial IIoT — Current Focus

- Automating shop-floor operations by integrating PLC and SCADA systems to enable real-time monitoring and control of manufacturing equipment.
- Designing IIoT data pipelines that acquire machine and sensor data from the shop floor and stream it to cloud services for storage, visualisation, and analysis.
- Developed a Manufacturing Execution System (MES) that connects shop-floor IIoT data to the enterprise ERP, closing the loop between machine-level operations and business systems for end-to-end OT/IT integration.
- Building the OT/IT data foundation required for predictive-maintenance and digital-twin initiatives, bridging factory-floor hardware with modern cloud and backend systems.
- Developing real-time dashboards that give operators and management live visibility into equipment status, throughput, and performance.
- Connected 25 CNC machines to a centralized IIoT platform, replacing manual data collection with automated, real-time data acquisition and monitoring.
- Automated data acquisition across three production lines — toughened-glass, gasket & rig, and PVC manufacturing — giving operators and management live visibility into machine status and production output.

Platform & Full-Stack Engineering

- Architected a multi-platform community-management ecosystem (web, Android, iOS) using Flutter and Dart, serving 5,000+ members with real-time communication features.
- Built scalable Node.js / MongoDB backend APIs handling 15,000+ daily requests at 99.9% uptime; optimised database schemas to improve query performance by 40%.
- Engineered an end-to-end influencer-marketing platform with a React analytics dashboard and brand-creator matching algorithms; reduced page load time by 35%.
- Led code reviews and mentored junior developers on design patterns and engineering best practices.

Software Developer

Mar 2014 – Feb 2022

Satyam Software Solutions Pvt. Ltd., Noida

- Built a local business-discovery platform (Just Dial-style) serving 5,000+ registered businesses using ASP.NET and MySQL.
- Designed and maintained the relational database architecture, advanced search / filtering, authentication, business-verification workflows, and rating/review systems.
- Partnered with cross-functional teams to gather requirements and deliver features on schedule.

FOCUS AREAS — CURRENTLY DEVELOPING

- **Predictive Maintenance:** applying machine learning to time-series sensor data to forecast equipment failures before they occur.
- **Digital Twin:** building virtual models of physical assets and processes for simulation, monitoring, and optimisation.
- **Industrial AI & Anomaly Detection:** detecting abnormal machine behaviour from streaming telemetry to improve reliability and quality.
- **MLOps foundations:** Python for data analysis, model training workflows, and deploying models against real-time industrial data streams.

EDUCATION

Master of Business Administration (MBA)

2011 – 2013

Uttar Pradesh Technical University, Lucknow

Master of Computer Applications (MCA)

2006 – 2009

Dr. Ram Manohar Lohia Avadh University, Ayodhya

Bachelor of Computer Applications (BCA)

2001 – 2004

Dr. Bhimrao Ambedkar University, Agra

CERTIFICATIONS & CONTINUING EDUCATION

- Web Development Specialization — Johns Hopkins University
- Introduction to Web Development (HTML, CSS, JavaScript) — IBM
- Fundamentals of Digital Marketing — Google Digital Garage
- Career Edge – Young Professional — TCS iON
- *[Add your in-progress Industry 4.0 / IIoT / AI-ML courses here — they strongly reinforce the pivot. See notes below.]*